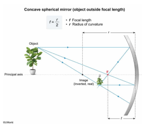
A concave mirror has a 4-m radius of curvature. This mirror will focus distant objects at a location that is approximately:

- A. 4 m in front of the mirror. [16%]
- B. 4 m behind the mirror. [8%]
- ✔ C. 2 m in front of the mirror. [61%]
- D. 2 m behind the mirror. [12%]

Correct
62% answered correctly

Explanation:



Spherical mirrors are a specific class of curved reflective surfaces shaped like a portion of a sphere. Spherical mirrors are further classified as **concave (converging)** or convex (diverging). The reflective surface of concave (converging) mirrors curves inward, away from the object and incident (incoming) light. Conversely, the reflective surface of convex (diverging) mirrors curves outward, toward the object and incident light.

The **focal length (*f*)** of a concave mirror is the distance in front of the mirror at which reflected light rays converge to form an image. Concave mirrors will form an **inverted**, **real image** when the reflected object is located outside of the focal length, but when the object is placed inside the focal length of a concave mirror, an upright, virtual image is formed. In contrast, convex mirrors always form an upright, virtual image.

Ignoring spherical aberrations, the focal length of a spherical mirror is equal to *half* the radius of curvature (*r*) **(Choice A)**, where *r* is the distance between the midpoint of the mirror surface and the center of the imaginary sphere of which the mirror surface is only a portion:

$$f = \frac{r}{2}$$

The equation for focal length is identical for both convex and concave mirrors. Consequently, a concave mirror and a convex mirror with the same radius of curvature will have identical focal length magnitudes. However, the point of focus is positioned *in front* of a concave mirror and *behind* a convex mirror **(Choices B and D)**.

This question introduces a concave mirror with a radius of curvature of 4 m. The focal length of the concave mirror may be calculated as:

$$f = \frac{4\ \text{m}}{2} = 2\ \text{m}$$

Because concave mirrors converge distant objects to form real, inverted images beyond the focal length of the mirror, this concave mirror will focus the light rays emanating from a distant object at a location that is at least 2 m in front of the mirror.

Educational objective:
Concave (converging) spherical mirrors create real, inverted images of objects placed outside of mirror focal length. The focal length of a concave spherical mirror is half the radius of curvature and is positioned in front of the mirror.

Time spent:0 sec
Subject:
Physics

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System:
Light and Sound